

ComfyUI

Deep Dive Into Node-Based Image Generation

CAS AICP Module 4 Exercise
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ComfyUI

A node-based interface to run diffusion models

Exercise questions

- (1) *What is ComfyUI, how does it work and what are its strengths and weaknesses?*
- (2) *Is ComfyUI the right tool for my project, a fully AI generated short film?*
- (3) *Can I generate a first (reference) frame for my project?*

Key milestones

- Install and run ComfyUI
- Work through basic tutorials
- Initial attempts at generating the first frame of my AICP film project
- Explore manipulations and variations

Project Sneak Peak

AI generated short film about the «Protection Gap»

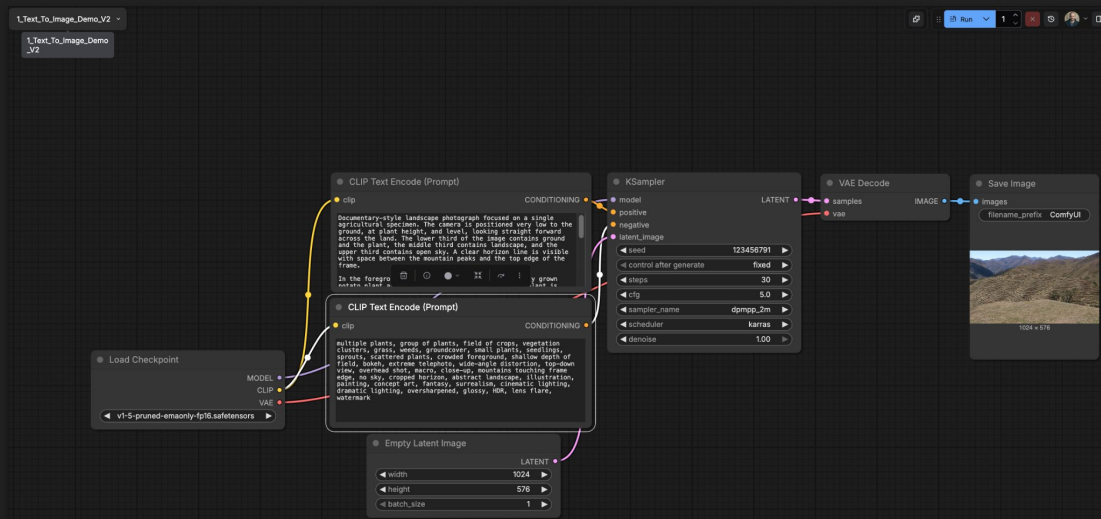
Conceptual framework

- A visual journey across three eras, social systems and aspects of the protection gap:
 - 1: Andes, Inca Empire, pre-collapse (1450–1525)
 - 2: Switzerland, Einsiedeln Region, pre-flooding of Lake Sihl (1925-1935)
 - 3: Mars, speculative future agriculture (2150-2200)
- Main Character: The potato (plant)
- First composition: Chapter 1



Text to image (1)

Workflow for generating reference frames



Here's an explanation of the KSampler node parameters:

Parameter Name	Description	Function
model	Diffusion model used for denoising	Determines the style and quality of generated images
positive	Positive prompt condition encoding	Guides generation to include specified elements
negative	Negative prompt condition encoding	Suppresses unwanted content
latent_image	Latent space image to be denoised	Serves as the input carrier for noise initialization
seed	Random seed for noise generation	Controls generation randomness
control_after_generate	Seed control mode after generation	Determines seed variation pattern in batch generation
steps	Number of denoising iterations	More steps mean finer details but longer processing time
cfg	Classifier-free guidance scale	Controls prompt constraint strength (too high leads to overfitting)
sampler_name	Sampling algorithm name	Determines the mathematical method for denoising path
scheduler	Scheduler type	Controls noise decay rate and step size allocation
denoise	Denoising strength coefficient	Controls noise strength added to latent space, 0.0 preserves original input features, 1.0 is complete noise

KSampler Components

Text to image (2): First generation

Landscapes, perspectives using a “documentary style”



Workflow for changing details

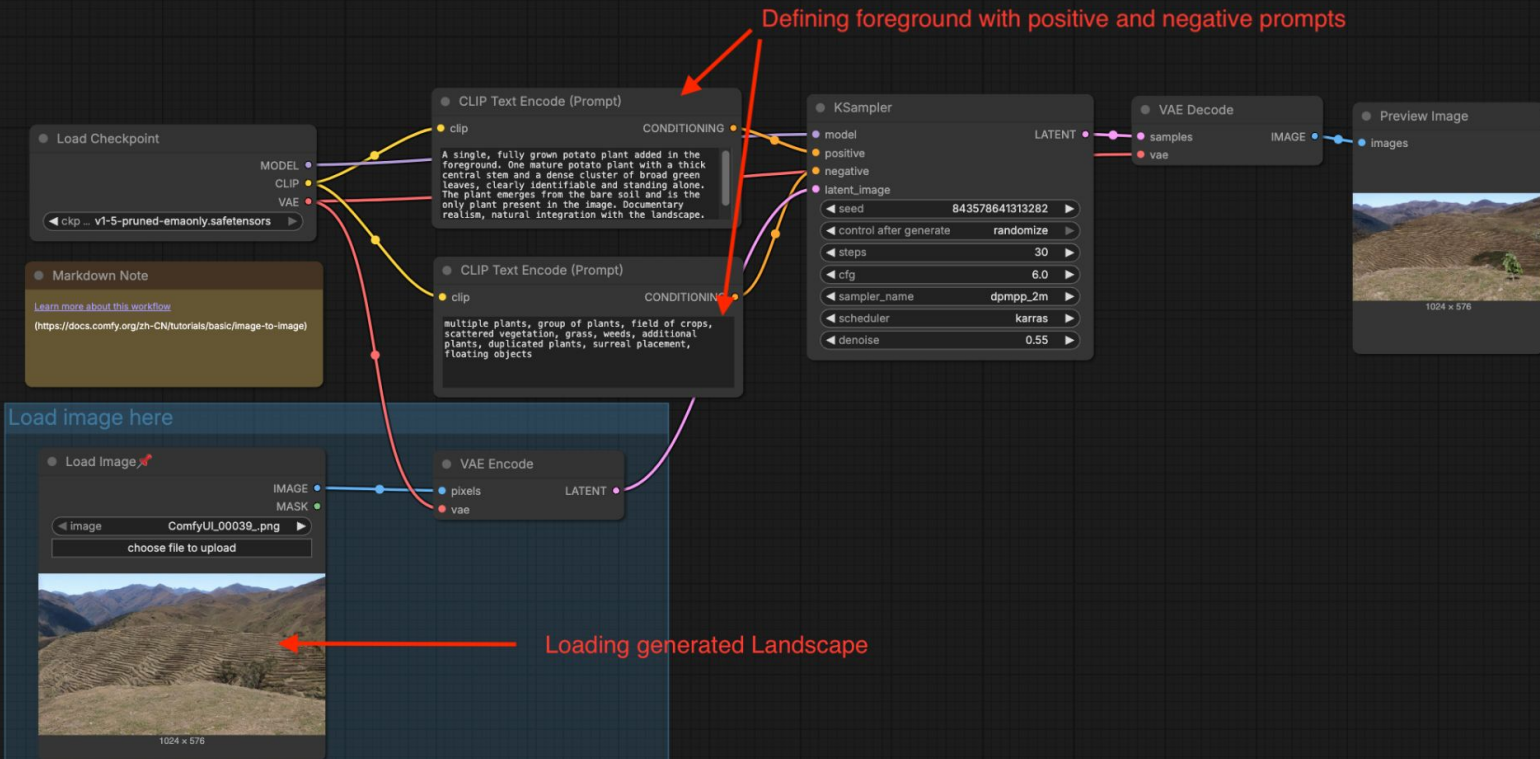
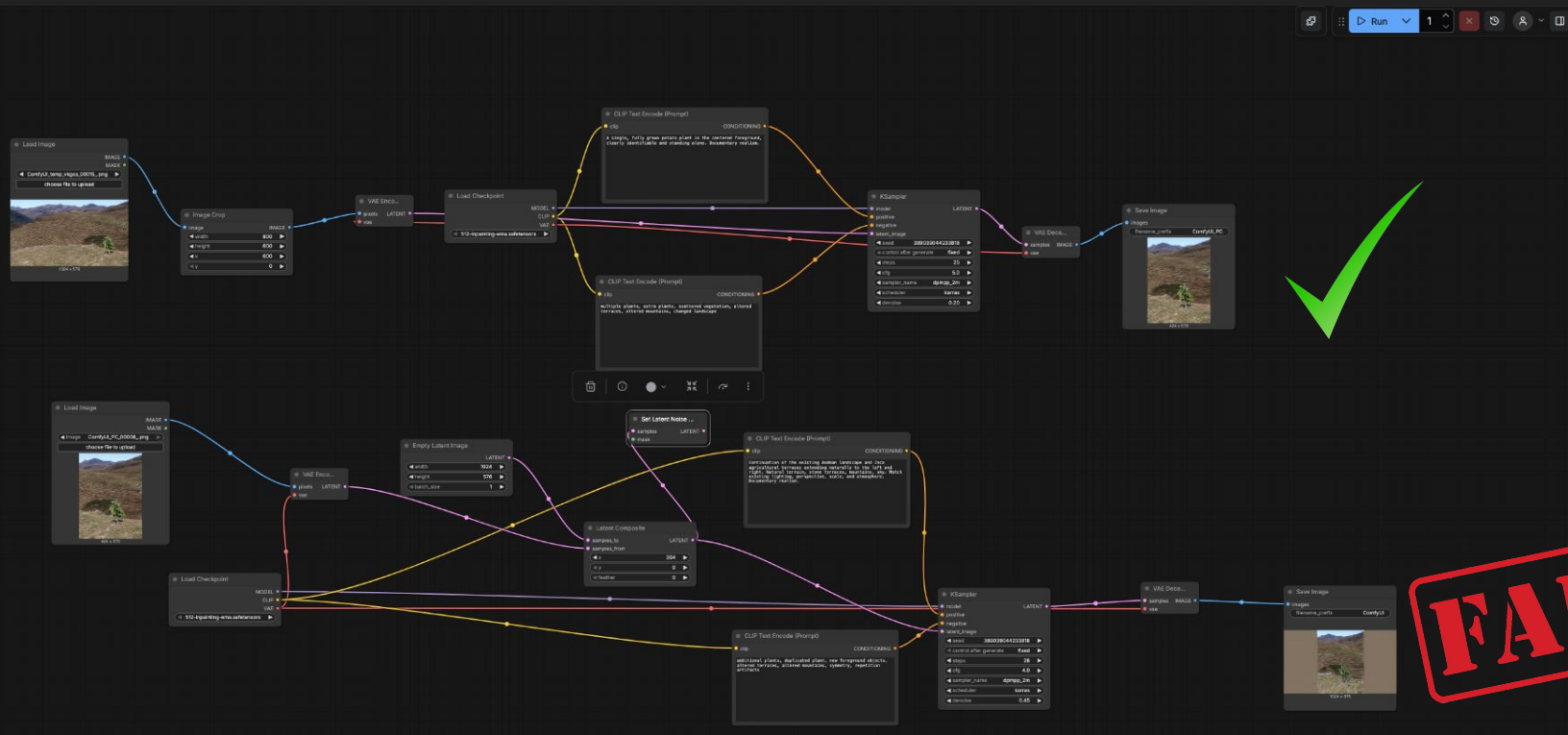


Image to image (2): Refined Generation

Adding a potato pant...



Reframing, Cropping & Extending Building Workflows with ChatGPT



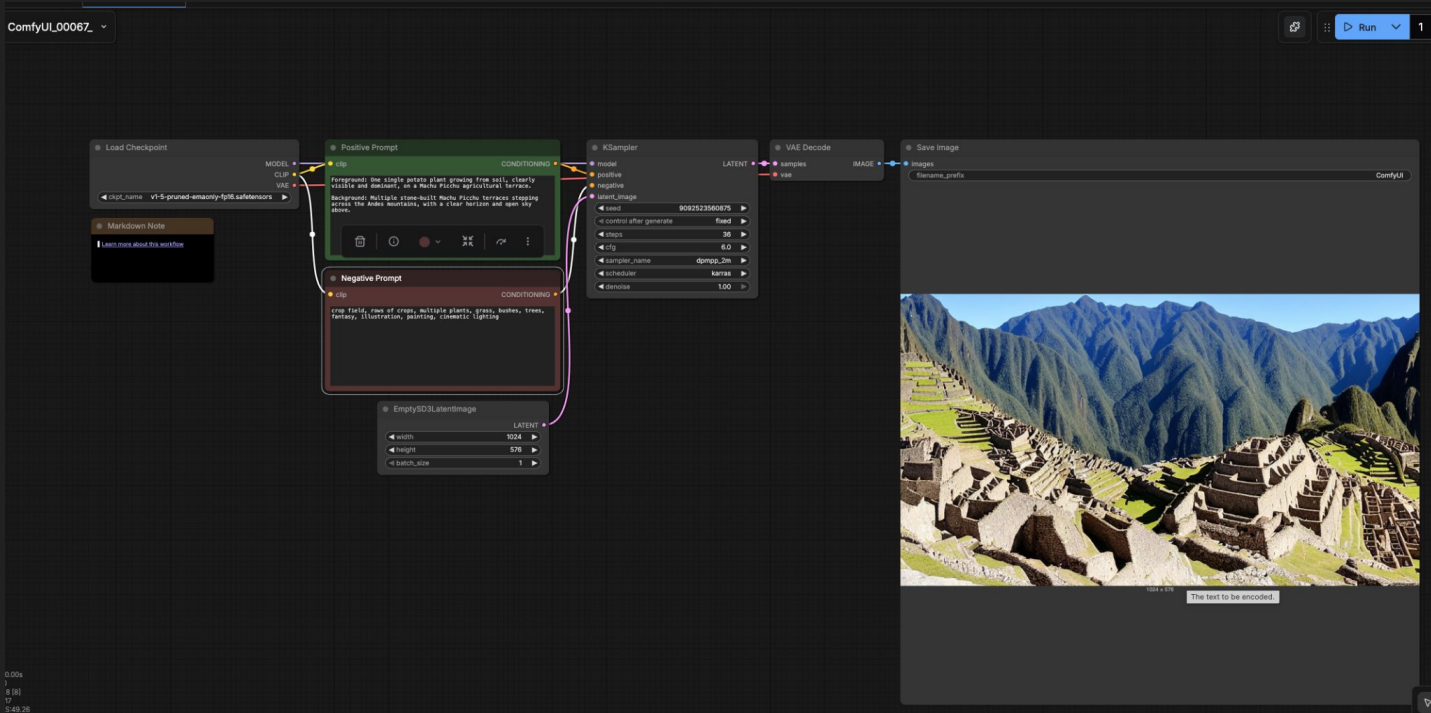
Resizing:

Bringing the centered image back to 16x9 format



Using StableDiffusion 3.5 (1)

ComfyUI Workflow template



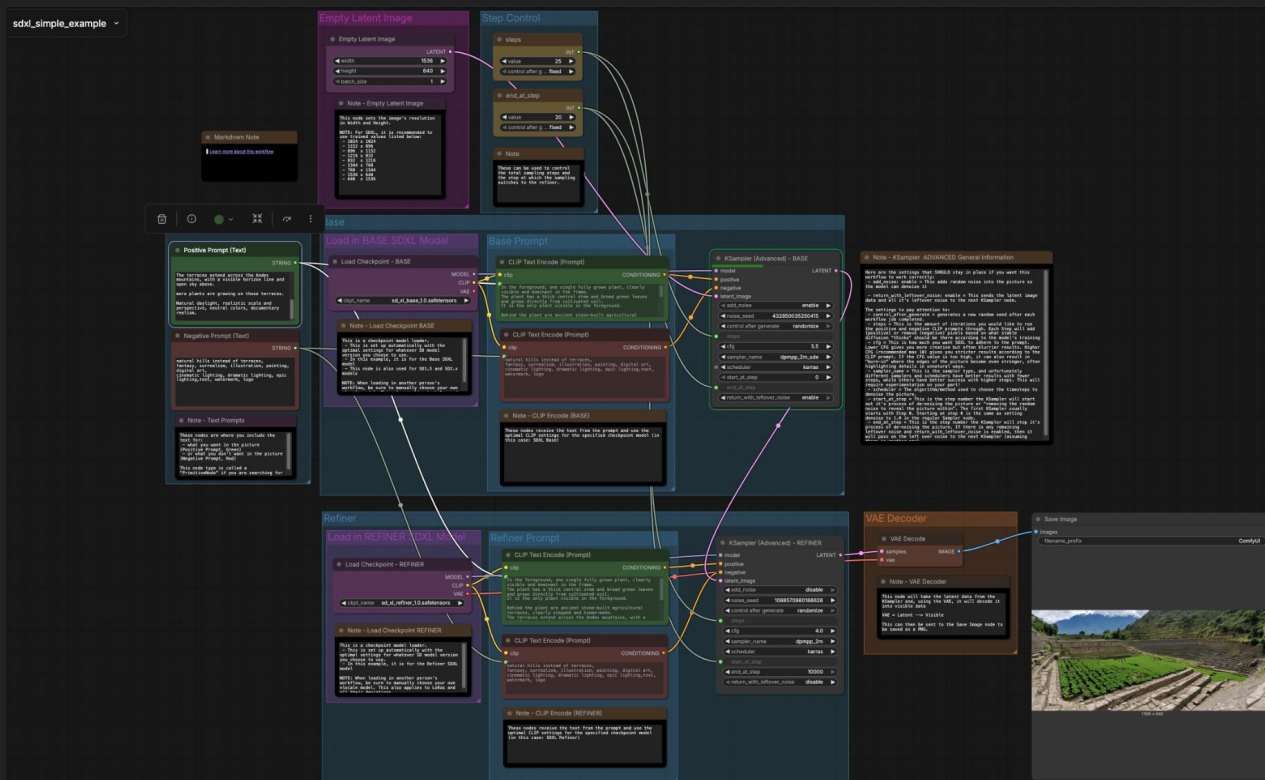
Using StableDiffusion 3.5 (2)

The difficulty of prompting images using raw diffusion



Stable Diffusion XL (1)

...to the rescue?



Stable Diffusion XL (2)



GPT orchestration vs. raw diffusion

Chat GPT outcome (one single prompt)

